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Biotransformation of fluorotelomer compound to perfluorocarboxylates in humans ☆

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ABSTRACT

Levels of perfluorocarboxylates (PFCAs) in biological compartments have been known for some time but their transport routes and distribution patterns are not properly elucidated. The opinions diverge whether the exposure of the general population occurs indirect through precursors or direct via PFCAs. Previous results showed that ski wax technicians are exposed to levels up to 92 000 ng/m^3 of 8:2 fluorotelomer alcohol (FTOH) via air and have elevated blood levels of PFCAs. Blood samples were collected in 2007–2011 and analyzed for C_4–C_{18} PFCAs, 6:2, 8:2 and 10:2 unsaturated fluorotelomer acids (FTUCAs) and 3:3, 5:3 and 7:3 fluorotelomer acids (FTCAs) using UPLC–MS/MS. Perfluorooctanoic acid (PFOA) was detected in levels ranging from 1.90 to 628 ng/mL whole blood (wb). Metabolic intermediates 5:3 and 7:3 FTCA were detected in all samples at levels up to 6.1 and 3.9 ng/mL wb. 6:2, 8:2 and 10:2 FTUCAs showed maximum levels of 0.07, 0.64 and 0.11 ng/mL wb. Also, for the first time levels of PFHxDA and PFOcDA were detected in the human blood at mean concentrations up to 4.22 ng/mL wb and 4.25 ng/mL wb respectively.
The aim of this study was to determine concentrations of PFCAs and FTOH metabolites in blood from ski wax technicians.

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1. Introduction

Per- and polyfluorinated compounds (PFASs) were used in commercial products already in the 50s because of their chemical and thermal resistance and ability to repel both oil and water (Speel and Schwarz, 1957). Today this diverse group of chemicals are extensively used in textile treatment, food packaging paper, stain repellant, hydraulic oil, non-stick cooking pans, ski wax and several other consumer products (Plassman, 2010; Prevedouros et al., 2006). Paradoxically, the thermal and chemical persistence which makes PFASs ideal in many industrial applications is also what makes them problematic in the environment due to their long half-lives. As a result, many PFASs have been found in humans, oceans, air and wildlife from all over the world (Butt et al., 2010a; Jahnke et al., 2007; Kärrman et al., 2005; Martin et al., 2004; Rotander et al., 2012). Around the millennium scientists became aware of the adverse effects in animals exposed to perfluorooctane sulfonate (PFOS), a notorious member of the PFAS family, including cancer, reduced fertility, suppression of immune system, hormonal alteration and changes in fatty acid metabolism (Review by Lau et al.) (Lau et al., 2007). PFOS was detected in wildlife worldwide and evidence grew that it has a strong tendency to accumulate in human and animal tissues and could potentially pose a risk to human health and the environment over the long term (3M, 2000a; Giesy and Kannan, 2001; Olsen et al., 1999; Sohlenius et al., 1993). This made 3M, a major North American manufacturer phase out PFOS and PFOS related products in May 2000 (3M, 2000b). In 2009 PFOS was added to the United Nations list of persistent organic pollutants, the Stockholm Convention of persistent organic pollutants (POPs) meaning that it is regulated in many countries.

Another member of the PFAS family, perfluorooctanoic acid (PFOA), is not yet on the Stockholm Convention List but in 2006 the U.S. Environmental Protection Agency (EPA) and the eight major PFOA manufacturers launched the 2010/15 PFOA Stewardship Program in which the companies committed to eliminate emissions and product content of PFOA by 2015 (EPA, 2010). PFOA is associated with liver toxicity, immunotoxicity, fatty acid metabolism, and developmental toxicity in rodents and humans (Elcombe et al., 2010; Fei et al., 2007; Klaunig et al., 2011; Lau et al., 2006; Yang et al., 2000). Also, U.S. EPA considers in its risk assessment of PFOA exposure that it is likely carcinogenic to humans (EPA, 2006). Therefore it is important to elucidate the metabolic pathways of all PFASs that potentially degrade to PFOA including the fluorotelomer alcohols (FTOHs). Already in 1981 Hagen et al. reported biotransformation of FTOH to PFOA in rats. More recent research shows that 8:2 FTOH rapidly metabolizes to biopersistent PFCAs in rats, mice and fish. However, the international literature lacks data on whether this metabolization can also occur in humans (Brandsma et al., 2011; Butt et al., 2010b; Fasano et al., 2006; Hagen et al., 1981).

☆ Brief: The study suggests that fluorotelomer alcohols (FTOHs) biotransform to perfluorocarboxylates in humans implying that evaluation of perfluorooctanoic acid (PFOA) exposure to humans should include FTOH exposure.

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Fluorochemicals are used in ski wax gliders since they reduce the friction between the ski and the snow consequently generating a higher speed (Swixsport, 2012). In our earlier studies on ski wax technicians' PFAS exposure we confirmed high levels of PFOA in the blood in addition to very high levels of FTOHs in the air during ski waxing (Nilsson et al., 2010a, 2010b). In this paper we aim to report the PFAS blood levels of the same technicians included in previous studies during 4 consecutive years, '07–11'. We also investigate the presence of metabolites FTCAs and FTUCAs in the blood previously identified as degradation products from FTOH exposure.

2. Materials and methods

2.1. Study design

We collected blood samples ($n = 94$) from 11 male professional ski wax technicians at 11 World Cup events in Cross Country skiing 2007–2011 at the end of the working day and monthly from April–August 2008 when the technicians were not working with ski wax. However, the technicians use fluorinated ski wax for approximately 30 h a week during the skiing season from December through March. The technicians' length and weight are presented in Table A.5 in the Appendix.

Before the study we received written informed consent from all participants and the ethical vetting board of Uppsala, Sweden (Reference No. Dnr 2010/ 056) approved of the study protocol.

2.2. Chemicals

Details of chemicals, abbreviations of PFASs and standard compounds used for the sample preparation are given in the Appendix on page A3 and A4. The molecular structures are shown in Fig. 1.

2.3. Extraction and analysis

Detailed information of the sample extraction has been described elsewhere (Kärrman et al., 2007). In brief, mass labeled internal standards (IS) and 2 mL of formic acid/water (1:1 v/v) were added to 0.5 mL of whole blood. After sonication and centrifugation, the supernatant was extracted using solid-phase extraction with weak anion exchange (Waters Oasis WAX), and the per- and polyfluorinated compounds were eluted with 4 mL methanol and 1 mL of 2% ammonium hydroxide in methanol. The volume of the blood extract was adjusted to 200 μL using nitrogen. Recovery standard (RS) and 300 μL of 2 mM ammonium acetate in water were added prior to UPLC–MS/MS analysis.

Levels of C_4–C_{14}, C_{16} and C_{18} PFCAs and 3:3, 5:3 and 7:3 FTCAs and 6:2, 8:2 and 10:2 FTUCAs were monitored. C_4, C_6, C_8 and C_{10} PFSAs were also measured in the same method and were included in the study. However, PFSAs are not further discussed in the paper since earlier results showed that the ski wax technicians do not have elevated PFSA blood levels (Nilsson et al., 2010b). Concentrations of PFHxDA and PFOcDA were measured semi quantitative since no mass labeled standard are available for those compounds and therefore these results are not compensated for possible matrix effects. The metabolites were chosen based on previous works on mammalian biotransformation of FTOH (Fasano et al., 2009; Nabb et al., 2007). An Acquity UPLC system coupled to a Quattro Premier XE (Waters Corporation, Milford) with an atmospheric electrospray interface operating in negative ion mode (ES–MS/MS) was used for analysis. Separation was performed on an Acquity BEH C18 2.1 × 100 mm, 1.7 μm kept at 50 °C with a PFCs isolator column (Waters Corporation, Milford) inserted between the pump and injector to retain any fluorochemicals originating from the UPLC system. Injection volume was 10 μL and the flow was set to 300 μL/min.

MRM-transitions, mass labeled internal and recovery standards used for quantification of analytes are presented in Tables A.1 and A.2 in the Appendix.

2.4. QA/QC

A calibration curve at concentrations 0.04, 0.08, 0.2, 0.4, 0.8, 2.0, 4.0, 8.0, 20 and 40 ng/mL was used for C_4–C_{14}, C_{16} and C_{18} PFCAs, 6:2 FTUCA, 8:2 FTUCA, 10:2 FTUCA, 3:3 FTCA 5:3 FTCA and 7:3 FTCA. The relative response factors of all analytes over the whole range showed relative standard deviations (RSD) of less than 15% except for perfluorooctadecanoic acid (PFOcDA, C_{18}) from 0.08 to 4 ng/mL. Quantification was performed using the internal standard method. The limit of detection (LOD) was defined as mean concentration in blanks + 3 standard deviations (σ), and the limit of quantitation (LOQ) was defined as mean blank level + 10σ. All LODs and LOQs are shown in Table A.3 in the Appendix. Mean recoveries ($n = 94$) for labeled PFOA, 6:2 FTUCA and 10:2 FTUCA were 86%, 72% and 52% respectively. The reported levels were recovery corrected. Mass labeled FTCAs to be used as internal standards were not available therefore the 3:3, 5:3 and 7:3 FTCAs were quantified using the FTUCAs according to Table A.2 in the Appendix. Spiking experiments showed mean ($n = 6$) matrix recoveries for 3:3, 5:3 and 7:3 FTCAs of 56%, 88% and 89% which corresponds well with the FTUCAs. No FTUCAs or FTCAs were present in in-house reference blood samples (human whole blood) or blanks. Samples were stored in freezer prior to analysis. No contamination was detected from utensils and chemicals which were evaluated by analyzing multiple blanks and reference blood samples stored more than 2 years.

3. Results

PFOA, PFNA, PFDA, PFDoDA, PFTDA, 5:3 FTCA and 7:3 FTCA were detected in all samples. Their individual levels are presented in the Appendix, Table A.4. Also, levels of the long chained PFCAs, PFHxDA and PFOcDA were detected in the human blood at mean concentration of 1.09 ng/mL wb and up to 4.22 ng/mL wb for PFHxDA and 0.69 ng/mL wb and up to 4.25 ng/mL wb for PFOcDA.

Fig. 2 shows UPLC–MS/MS multiple reaction monitoring (MRM) chromatograms of polyfluorinated metabolites FTCAs and FTUCAs. 5:3 FTCA concentrations range between 0.09 and 6.1 ng/mL wb and 7:3 FTCA between 0.05 and 3.5 ng/mL wb. 3:3 FTCA was not detected in levels exceeding LOQ in any of the samples but 8:2 FTUCA and 10:2 FTUCA were detected in 84% and 39% of the samples in ranges of as presented in Tables 1and A.4 in the Appendix.

The technicians can be divided in two groups with regard to the trend in blood concentrations of PFOA. For the group with PFOA levels exceeding 100 ng/mL wb at the beginning of the sampling campaign in September 2007, the concentration of PFOA has decreased until the last sampling in March. The other group with PFOA levels less than 100 ng/mL wb has, generally, increased the blood concentrations from 2007 to 2011. PFNA levels range from 0.33 to 163 ng/mL wb with a mean of 23 ng/mL wb which is up to 300 times higher levels than those found in the general population (Haug et al., 2009; Kato et al., 2011; Toms et al., 2009).

4. Discussion

Our findings of FTOH metabolites FTCAs and FTUCAs in the blood from ski wax technicians exposed to high levels of 8:2 FTOH via air suggest metabolism of FTOHs to PFOA and PFNA also in humans (Fasano et al., 2009; Nabb et al., 2007). This data is important to explain biological levels of PFCAs because the parent compound itself may not accumulate but biotransformation forms persistent metabolites PFOA and PFNA. In fact, it is reported that the FTCA metabolites were responsible for toxic effects in rats and microorganisms rather than the parent compound FTOH (Martin et al., 2009; Phillips et al., 2007). In addition, we are able to report the blood level of PFHxDA and PFOcDA in wax technicians of concentrations up to 4.22 ng/mL blood for PFHxDA and up to 4.25 ng/mL PFOcDA.

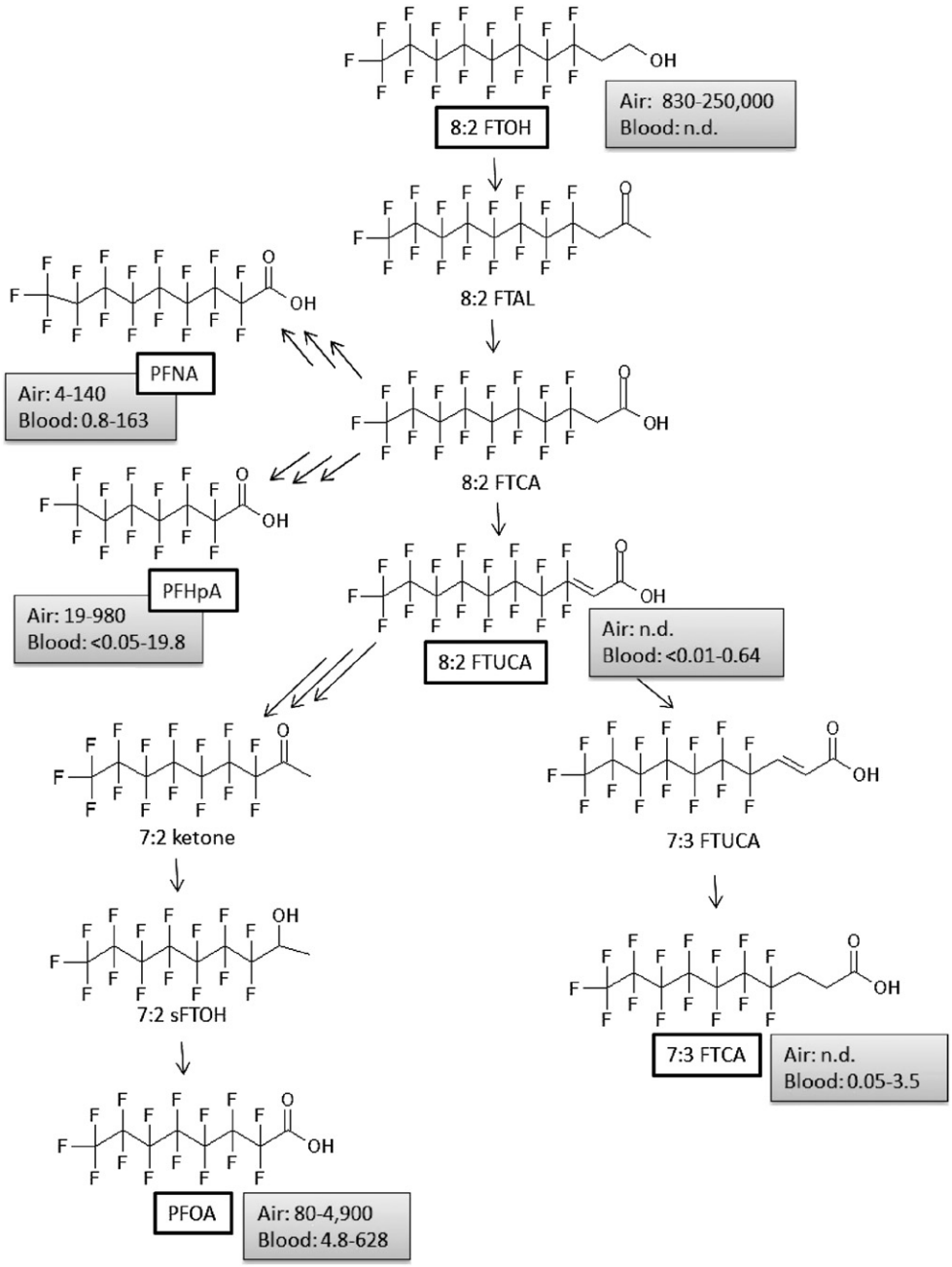

Fig. 1. Simplified metabolic pathway for 8:2 FTOH showing combined suggestions by Martin et al. (2005), Fasano et al. (2009), Martin et al. (2009) and Himmelstein et al. (2011). The ranges of blood and air levels of the compounds included in the pathway are presented in ng/mL for blood and ng/m^3 for air (Nilsson et al., 2010a, 2010b).

In Fig. 1 a simplified metabolic pathway for 8:2 FTOH metabolism in rat is presented by combining the findings from Fasano et al. (2006), Martin et al. (2005) and Nabb et al. (2007) including our results of PFHpA, PFOA, PFNA, 7:3 FTCA and 8:2 FTUCA levels in blood and air. The metabolic routes of FTOHs in in vivo rainbow trout and rat and in vitro human hepatic cells are very complex including phase I (oxidation)

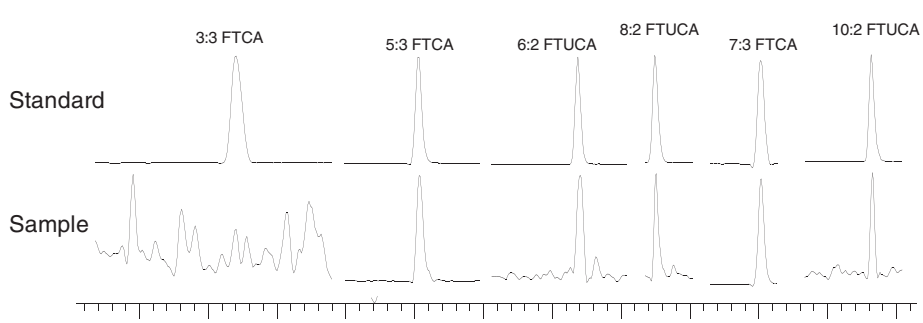

Fig. 2. UPLC–MS/MS multiple reaction monitoring chromatograms of polyfluorinated metabolites identified in the blood of humans exposed to extremely high levels of 8:2 FTOH via air.

Table 1
Minimum, maximum, median and average concentration (ng/mL whole blood) of PFASs in the blood from ski wax technicians ($n = 11$) collected 2007–2011.

		Min	Max	Median	Average
PFCAs	PFBA	<0.13	11	0.51	1.8
	PFPeA	<0.05	0.18	0.14	0.14
	PFHxA	<0.05	15	0.68	1.9
	PFHpA	<0.05	20	2.4	4.0
	PFOA	1.9	630	110	130
	PFNA	0.33	160	12	23
	PFDA	0.87	24	7.2	7.8
	PFUnDA	<0.16	2.8	1.0	1.0
	PFDoDA	0.21	2.8	1.3	1.4
	PFTrDA	<0.10	1.1	0.39	0.44
	PFTDA	0.07	5.2	1.2	1.5
	PFHxDA	<0.15	4.2	1.2	1.4
	PFOcDA	0.20	4.3	0.85	1.0
PFSAs	PFBS	<0.02	0.04	0.04	0.03
	PFHxS	<0.04	4.3	1.4	1.5
	PFOS	0.28	27	11	11
	PFDS	<0.02	<0.02	<0.02	<0.02
Metabolites	6:2 FTUCA	<0.01	0.07	0.03	0.03
	8:2 FTUCA	<0.01	0.64	0.07	0.10
	10:2 FTUCA	<0.02	0.11	0.05	0.06
	3:3 FTCA	<0.03	<0.03	<0.03	<0.03
	5:3 FTCA	0.09	6.1	1.7	1.9
	7:3 FTCA	0.05	3.9	0.92	1.0

and phase II pathways (conjugation of parent compounds and metabolites) (Fasano et al., 2009; Himmelstein et al., 2011; Nabb et al., 2007). A similar metabolization complexity is also expected in in vivo humans.

In our study the average concentrations of 6:2 FTOH, 8:2 FTOH, and 10:2 FTOH are extremely high (280 ng/m^3, 92,800 ng/m^3, and 370 ng/m^3) in the air of wax technicians' breathing zone during work applying ski wax. PFOA is also detected in the air but at almost 100 times lower in level (mean = of 1 200 ng/m^3) than in 8:2 FTOH. Average levels of PFNA are 30 ng/m^3 compared to 4 900 ng/m^3 for PFHxA which is detected at the highest level of all the PFCAs in air (Nilsson et al., 2010a). Typical indoor air concentrations of 8:2 FTOH are approximately 5 ng/m^3 (Haug et al., 2011). The general population have on average 1.5 ng/mL PFOA and 0.3 ng/mL PFNA in their blood compared to the technicians who show levels of up to 635 ng/mL PFOA and 163 ng/mL PFNA (Haug et al., 2009; Kato et al., 2011; Shoeib et al., 2011).

The results found in our study confirming high levels of PFOA in the blood can result from either direct air exposure of PFOA and/or from the metabolism of 8:2 FTOH. In the literature, experiments where rats inhaled 3 or 30 mg/m^3 it was shown that 1.6–2.1% of the dosed 8:2 FTOH was metabolized to PFOA (Himmelstein et al., 2011). The technicians' occupational exposure to PFCAs is much higher than the exposure from the diet and ambient indoor air and therefore those are assumed to be a minor contributor to their blood levels (Haug et al., 2011). Low levels of FTUCAs (<0.002–0.080 ng/g) were recently detected in indoor environment however FTCAs were not detected in the dust samples from the same households (Jogsten et al., 2012).

It is difficult to draw any conclusions about how much of the FTOH is metabolized to PFOA. PFOA has a half-life of 2.3–3.8 years and therefore the blood levels reflect the exposure from several years back in time (Bartell et al., 2010; Olsen et al., 2007). The air data in our study are single-point measurements providing information of that particular working day only and we have seen that the air exposure can differ by several orders of magnitude from one occasion to the next (Nilsson et al., 2010a). Another factor is that the exposure consists of a cocktail of PFASs as most animal studies where the aim is to elucidate the metabolic pathways and yields only one chemical is dosed at a time. However, this is a unique group of workers providing an exceptional opportunity to study people highly exposed to FTOHs. The extreme exposure makes it possible to detect metabolic products that in a less exposed group would remain undiscovered.

Out of the metabolic products included in our study the most biopersistent compounds in humans are the 5:3 and 7:3 FTCAs since they are detected in the samples collected throughout the summer even though the technicians are not working with ski wax during the summer. The levels of 5:3 and 7:3 FTCAs are shown in Table A.4 and Figs. A.1 and A.2 in the Appendix.

Butt et al. showed that biotransformation of a different fluorochemical, 8:2 fluorotelomer acrylate (FTAc), metabolizes via 8:2 FTOH in rainbow trout and yields 8:2 FTCA, 8:2 FTUCA and 7:3 FTCA as the most biopersistent metabolites (Butt et al., 2010b). Both 8:2 FTUCA and 7:3 FTCA were present at very low levels in our samples. However, 7:3 FTCA was detected in all of the blood samples although levels declined in summer samples. A biotransformation pathway of 8:2 FTOH in rat was reported by Fasano et al. and also there the 7:3 FTCA was identified as a major metabolite (Fasano et al., 2009). Generally, we observed slightly higher levels of 5:3 FTCA as compared to 7:3 FTCA but in 22% of the blood samples 7:3 FTCA levels were higher than 5:3 FTCA. Experiments with rats and rainbow trout have demonstrated that the 7:3 FTCA does not form PFOA but PFHxA (Butt et al., 2010b; Himmelstein et al., 2011).

Similar to this study, lower PFNA levels compared to PFOA were reported after 8:2 FTOH exposure in mammalian systems by several researchers (Butt et al., 2010b; Himmelstein et al., 2011; Kudo et al., 2005; Nabb et al., 2007). In our survey both PFOA and PFNA are present at high concentrations in the blood over a long period of time and show similar fluctuation indicating that they are persistent end products originating from a common source as exemplified in Figs. A.1 and A.2 in the Appendix. This is an unexpected finding since PFNA has a longer half-life than PFOA in rats and therefore is expected to eliminate more slowly also in humans (Ohmori et al., 2003; Tatum-Gibbs et al., 2011). Scientific reports after the 3M phase-out show decreasing PFOA serum concentrations in the general population, whereas levels of PFNA still increase (Calafat et al., 2007). However, increased background exposure is not likely the explanation for the high PFNA levels seen in wax technicians which are on average 50 times higher than for the general population (Haug et al., 2009; Kato et al., 2011).

Since the study began in 2007 three more technicians have accepted to participate in the study. One of these show the highest levels of PFOA with 628 ng/mL wb which corresponds to approximately 1256 ng/mL serum (Ehresman et al., 2007). What is also noteworthy is that most technicians have PFOA concentrations 7–10 times higher than PFNA but one of the new technicians show 37 times higher level of PFOA than PFNA. PFHxA has a half-life in the blood of less than one month since it is not present in the samples collected in April one month after the end of the World Cup ski season (Table A.4 in the Appendix). This is in agreement with PFHxA half-lives of 2.4–5.3 h in monkeys which has been reported previously (Chengelis et al., 2009).

5. Conclusion

The novel findings of FTOH metabolites FTCAs and FTUCAs in the blood from ski wax technicians exposed to high levels of 8:2 FTOH via air is indicating metabolism of FTOHs to PFOA and PFNA also in humans (Fasano et al., 2009; Nabb et al., 2007). Our data may be important in the discussion of PFOA phase-out since it has been reported that the general population is exposed to FTOHs in air and dust (Dreyer et al., 2009; Shoeib et al., 2011).

In addition, for the first time we present the levels of PFHxDA and PFOcDA in the human blood.

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Appendix A. Supplementary data

Available in Appendix are the following: List of abbreviations, chemicals, UPLC gradient program, common name, acronym, LC/MS–MS MRM transitions of analytes, limit of detection (LOD) and limit of quantification (LOQ) for the analytes, individual concentration (ng/mL whole blood) of per-/polyfluorinated compounds in the blood from ski wax technicians ($n = 11$) collected in 2007–2011.

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.envint.2012.09.001.

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